BIOS VS UEFI: A COMPARISON

ITEP 414 - System Administration and Maintenance | Week 3

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Comparison Table

Aspect	BIOS	UEFI
Definition	Legacy firmware that has been around since the first IBM PCs in the 1970s. It lives on a small chip on the motherboard and its only job is to get the hardware ready before the operating system takes over.	Modern firmware standard managed by the UEFI Forum. It is basically a small operating system that sits between the hardware and the OS, with its own drivers and a much richer interface.
Boot Process	Checks the first 512 bytes of the disk (the Master Boot Record) and jumps straight into whatever boot code is written there. One step at a time, no shortcuts.	Reads boot entries from a dedicated EFI System Partition and loads boot managers directly, like GRUB or the Windows Boot Manager. It can also initialize several devices at the same time.
Maximum Disk Support	2 TB. Once a disk is bigger than that, BIOS simply cannot see the rest of it.	Practically unlimited, up to 9.4 ZB. Large disks are not a problem because it uses 64-bit addressing.
Partition Style	MBR, which only allows 4 primary partitions per disk.	GPT, which allows up to 128 partitions on one disk.
Security Features	Almost none. It blindly executes whatever is in the boot sector, which is why bootkits were a thing.	Secure Boot. Every boot loader and kernel is checked against trusted certificates before it is allowed to run.
Speed	Slower. It initializes hardware one component at a time and only works in a 16-bit mode.	Faster. Devices are detected in parallel and the firmware uses 32-bit or 64-bit modern drivers.
Advantages	Works with very old hardware and old 32-bit operating systems. Simple enough that most people understand it.	Big disk support, Secure Boot, faster startup, a graphical setup screen, network boot, and drivers built into the firmware itself.
Disadvantages	2 TB limit, 4 partition limit, no security checks, slow boot, and a design that has not changed in 40 years.	Requires GPT disks and an OS that understands UEFI. Old machines cannot run it, and the setup options can be confusing at first.
Modern Usage	Practically retired. It survives only as a legacy or compatibility mode for old systems.	Standard on every modern PC, laptop, and server since around 2012, and required for Windows 8 and later.

Why UEFI Has Replaced BIOS

When I was installing Ubuntu Server in VirtualBox for this project, the virtual machine settings asked me to choose between BIOS and UEFI as the firmware type. It was the first time I really stopped to think about what that choice meant, so I looked into it. The short version is that BIOS was designed in the 1970s, and the world of computers has simply moved past it.

The easiest problem to understand is disk size. BIOS can only read up to 2 TB of a disk, because its address system is 32-bit and works with the old MBR partition table. Modern drives are already bigger than that, and servers that store databases and file shares for a company will definitely need more space someday. UEFI uses the GPT partition table with 64-bit addressing, so a 2 TB limit simply does not exist anymore. It also allows up to 128 partitions instead of 4, which matters when one server runs several services, as our Ubuntu Server eventually will.

Security is the bigger reason, in my opinion. With BIOS, the computer executes whatever code is sitting in the first 512 bytes of the disk without checking where it came from. That is how bootkits and early malware like Stuxnet managed to hide deep in the system. UEFI introduced Secure Boot, which verifies the signature of the boot loader and the kernel against certificates stored in the firmware before anything runs. For a company server, that is a meaningful layer of protection that starts before the operating system is even loaded.

There is also speed and convenience. UEFI initializes hardware in parallel, so the machine reaches the login prompt noticeably faster. The firmware itself has drivers, a mouse-friendly setup interface, network booting, and even a scripting environment. It is a small operating system on its own, and it can be updated and extended, while BIOS is fixed code that nobody can really improve.

The industry made the decision for us anyway. Microsoft required UEFI with Secure Boot for Windows 8 certification in 2012, and every motherboard and laptop vendor followed. Today, almost everything I work with, including the virtual machine in this project, boots through UEFI. BIOS still shows up as a compatibility option for old software, but it is no longer the standard. For capacity, for security, and for speed, UEFI is simply what makes sense now.

References

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